



REPUBLIC OF THE PHILIPPINES

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
COLLEGE OF COMPUTER AND INFORMATION SCIENCES



CERTIFICATION OF VALIDATION

Date: September 11, 2025

To Whom It May Concern,

This is to formally certify that I, Kyle Christopher L. Melchor, an Artificial Intelligence and Machine Learning Engineer, have reviewed and validated the dataset utilized in the study entitled: **“SettleNet: ConvNext encoder-decoder semantic segmentation of Irregular High-Density Settlements with multimodal fusion RGB Satellite, Building Count, and Height maps”**.

Furthermore, I have assessed and validated the consistency, uniformity, and integrity of the data which was used in the study. This includes the satellite imagery, building density count, and building height dataset from both the Maxar Open Data and Google Open Buildings 2.5D Temporal sources. I deem the sources fit for the methodology the research has currently undertaken and will undertake. It should be noted that all work should be aligned to a Binary Semantic Segmentation approach for the categorization of pixels into irregular high-density settlements or informal settlements.

I have reviewed and verified the licenses connected to the proper use of the aforementioned datasets. Sources for the licenses and a short summary of its proper use are also given below. The licenses are as follows:

1) Maxar Open Data for Typhoon Vamco has a Creative Commons Attribution License for Non Commercial use. This allows the researchers to share and adapt the dataset as they see fit. However, the researchers must give appropriate credit, provide a link to the license, and indicate if changes were made to the dataset they used. Any reasonable manner to convey this information is possible, but it should never indicate that Maxar Open Data endorses the researchers or the research in any way. The full Creative Commons License can be found here.

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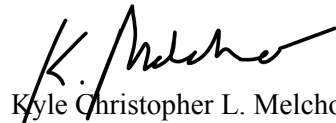
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I acknowledge the accuracy, quality, and relevance of the dataset and their appropriate licenses for use in this academic research on **“SettleNet: ConvNext encoder-decoder semantic segmentation of Irregular High-Density Settlements with multimodal fusion RGB Satellite, Building Count, and Height maps”**.

Should further verification be required, I can be reached via email at melchor.kyle@gmail.com.



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